

Weiming Feng

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🌐 <https://fwm94.github.io/>

Work Experience

Assistant Professor

Feb 2025 - Now

School of Computing and Data Science, The University of Hong Kong.

Junior Fellow

Jan 2024 - Jan 2025

Institute for Theoretical Studies, ETH Zürich

Simons–Berkeley Research Fellow

Aug 2023 - Dec 2023

Simons Institute for the Theory of Computing, UC Berkeley

Research Associate

Oct 2021 - Sep 2023

School of Informatics, The University of Edinburgh

Education

Ph.D. in Computer Science

Sep 2016 - June 2021

Field: Theoretical Computer Science

Advisor: Professor Yitong Yin

Nanjing University

B.Eng. in Network Engineering

Sep 2012 - June 2016

University of Electronic Science and Technology of China

Research Interests

Theoretical Computer Science:

- sampling and counting algorithms;
- discrete probability;
- learning theory.

Honors

CCF Outstanding Doctoral Dissertation Award (2021)

China Computer Federation

National Scholarship (2019)

MOE of PRC

Microsoft Research Asia Fellowship (2018)

MSRA

Gold Medals in ACM-ICPC Asia Regional Contests (2014-2015)

ACM

External Grants

Sampling and Approximate Counting for Undirected Graphical Models via Influence Bounds

HK\$ 792,146

Early Career Scheme, Research Grants Council of Hong Kong, 2025–2028.

Project number: 27202725.

Advisees

- Minji Yang (Ph.D. Student, 2024 –)
- Xiongxin Yang (Research Assistant, May 2025 - July 2025)
- Yixiao Yu (Research Assistant, May 2025 - July 2025)
- Yucheng Fu (Student Research Assistant, July 2025 - August 2025)

Publications

1. Weiming Feng, Heng Guo, and Minji Yang.
Rapid mixing in positively weighted restricted Boltzmann machines.
In **FOCS** 2026 (to appear).
2. Weiming Feng, Yucheng Fu, Minji Yang, and Anqi Zhang.
On computing total variation distance between mixtures of product distributions.
In **RANDOM** 2026 (to appear).
3. Weiming Feng, Zelin Li, and Pan Peng.
Sublinear-time algorithms for diagonally dominant systems and applications to the Friedkin-Johnsen model.
In **COCOON** 2026 (to appear).
4. Weiming Feng, Xiongxin Yang, Yixiao Yu, and Yiyao Zhang.
Learning CNF formulas from uniform random solutions in the local lemma regime
In **STOC** 2026: 478-488.
5. Weiming Feng and Yucheng Fu.
On approximating the f -divergence between two Ising models.
In **ITCS** 2026: 59:1-59:23.
6. Weiming Feng and Minji Yang.
Rapid mixing of Glauber dynamics for monotone systems via entropic independence.
In **SODA** 2026: 4894-4929.
7. Xiaoyu Chen, Weiming Feng, Zhe Ju, Tianshun Miao, Yitong Yin, and Xinyuan Zhang.
Faster mixing of the Jerrum-Sinclair chain.
In **FOCS** 2025: 1007-1028.
8. Xiaoyu Chen, Weiming Feng, Heng Guo, Xinyuan Zhang, and Zongrui Zou.
Deterministic counting from coupling independence.
In **FOCS** 2025: 667-683.
9. Xiaoyu Chen and Weiming Feng.
Rapid mixing via coupling independence for spin systems with unbounded degree.
In **RANDOM** 2025: 68:1-68:17.
10. Weiming Feng, Hongyang Liu, and Minji Yang.
Approximating the total variation distance between spin systems.
In **COLT** 2025: 1974-2025.
11. Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, Mark Jerrum, and Jiaheng Wang.
Rapid mixing of the flip chain over non-crossing spanning trees.
In **SoCG** 2025: 8:1-8:18.

12. Arnab Bhattacharyya, Weiming Feng, and Piyush Srivastava.
Approximating the total variation distance between Gaussians.
In **AISTATS 2025**: 1846-1854.
13. Weiming Feng and Ce Jin.
Approximately counting knapsack solutions in subquadratic time.
In **SODA 2025**: 1094-1135.
14. Charlie Carlson, Xiaoyu Chen, Weiming Feng, and Eric Vigoda.
Optimal mixing for randomly sampling edge colorings on trees down to the max degree.
In **SODA 2025**: 5418-5433.
15. Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, and Jiaheng Wang.
Approximate counting for spin systems in sub-quadratic time.
TheoretCS, Volume 4 (2025), Article 3, 1-27
Conference version in **ICALP 2024**: 11:1-11:20.
16. Weiming Feng and Heng Guo.
An FPRAS for two terminal reliability in directed acyclic graphs.
In **ICALP 2024**: 62:1-62:19.
17. Weiming Feng, Tianren Liu, and Liqiang Liu.
On deterministically approximating total variation distance.
In **SODA 2024**: 1766-1791.
18. Weiming Feng, Heng Guo, Chunyang Wang, Jiaheng Wang, and Yitong Yin.
Towards derandomising Markov chain Monte Carlo.
SIAM Journal on Computing 54(3), 775–813, 2025.
In **FOCS 2023**: 1963-1990.
19. Weiming Feng, Jiaheng Wang, and Heng Guo.
Swendsen-Wang dynamics for the ferromagnetic Ising model with external fields.
Information and Computation 294: 105066 (2023).
20. Charilaos Efthymiou and Weiming Feng.
On the mixing time of Glauber dynamics for the hard-core and related models on $G(n, d/n)$
In **ICALP 2023**: 54:1-54:17.
21. Weiming Feng, Heng Guo, Mark Jerrum, and Jiaheng Wang.
A simple polynomial-time approximation algorithm for the total variation distance between two product distributions.
TheoretCS Volume 2 (2023), Article 8, 1-7.
Conference version in **SOSA 2023**: 343-347.
22. Xiaoyu Chen, Weiming Feng, Yitong Yin, and Xinyuan Zhang.
Optimal mixing for two-state anti-ferromagnetic spin systems.
In **FOCS 2022**: 588-599.
23. Weiming Feng, Heng Guo, and Jiaheng Wang.
Improved bounds for randomly colouring simple hypergraphs.
In **RANDOM 2022**: volume 245 of LIPIcs, pages 25:1–25:17.
24. Weiming Feng, Heng Guo, Yitong Yin, and Chihao Zhang.
Rapid mixing from spectral independence beyond the Boolean domain.
ACM Transactions on Algorithms 18(3): 28:1-28:32 (2022).
Conference version in **SODA 2021**: 1558-1577.

25. Weiming Feng, Heng Guo, and Yitong Yin.
Perfect sampling from spatial mixing.
Random Structures & Algorithms 61(4): 678-709 (2022).
26. Xiaoyu Chen, Weiming Feng, Yitong Yin, and Xinyuan Zhang.
Rapid mixing of Glauber dynamics via spectral independence for all degrees.
SIAM Journal on Computing special section FOCS 2021.
Conference version in **FOCS 2021**: 137-148.
27. Weiming Feng, Heng Guo, Yitong Yin, and Chihao Zhang.
Fast sampling and counting k -SAT solutions in the local lemma regime.
Journal of the ACM 68 (6), 1-42, 2021.
Conference version in **STOC 2020**: 854-867.
28. Weiming Feng, Nisheeth K. Vishnoi, and Yitong Yin.
Dynamic sampling from graphical models.
SIAM Journal on Computing, 50(2), 350–381, 2021.
In **STOC 2019**: 1070-1081.
29. Weiming Feng, Kun He, and Yitong Yin.
Sampling constraint satisfaction solutions in the local lemma regime.
In **STOC 2021**: 1565-1578.
30. Weiming Feng, Thomas P. Hayes, and Yitong Yin.
Distributed Metropolis sampler with optimal parallelism.
In **SODA 2021**: 2121-2140.
31. Weiming Feng, Kun He, Xiaoming Sun, and Yitong Yin.
Dynamic inference in probabilistic graphical models.
In **ITCS 2021**: 25:1-25:20.
32. Weiming Feng, Yuxin Sun, and Yitong Yin.
What can be sampled locally?
Distributed Computing, 33, 227–253. 2020.
Conference version in **PODC 2017**: 121-130.
33. Weiming Feng and Yitong Yin.
On local distributed sampling and counting.
In **PODC 2018**: 189-198.

PREPRINTS

34. Weiming Feng, Xiongxin Yang, Yixiao Yu, and Yiyao Zhang.
Learning AC^0 under locally sampleable graphical models.
CoRR abs/2607.08303 (2026).
35. Weiming Feng, Heng Guo, and Yichun Yang.
Fast counting and sampling for ferromagnetic two-spin systems.
CoRR abs/2607.05248 (2026).

Talks and Posters

Recent advances in approximating f -divergences between two Ising models

Seminar Talk, Nanyang Technological University, Singapore, 2025

Seminar Talk, National University of Singapore, Singapore, 2025

Faster mixing of the Jerrum-Sinclair chain

ISBA Satellite Meeting on Information Geometry, Privacy and Monte Carlo, Institute of Statistical Mathematics,
Tokyo, Japan, 2026

Conference on Mixing Times between Probability, Computer Science and Statistical Physics, ICTP, Trieste, Italy, 2025

Mini-Course: Mixing time analysis for MCMC algorithms

University of Science and Technology of China, Hefei, China, 2025

An FPRAS for two terminal reliability in directed acyclic graphs

Summer School on Algorithms, Dynamics, and Information Flow in Networks, Möhnesee, Germany, 2024

ICALP, Tallinn, Estonia, 2024

On deterministically approximating total variation distance

SODA, Washington D.C., US, 2024

Sampling and approximately counting CNF formula solutions in the local lemma regime

Meet the Fellows Workshop, Simons Institute, Berkeley, California, US, 2023

Approximating the TV distance between two product distributions

MCMC 2.0 Workshop, NII Shonan Meeting, Shonan, Japan, 2023

Probabilistic Circuits and Logic Workshop, Simons Institute, Berkeley, California, US, 2023

On the mixing time of Glauber dynamics for the hard-core and related models on $G(n, d/n)$

ICALP 2023, Paderborn, Germany 2023

Sampling and counting hypergraph colourings

FODSI Workshop, Massachusetts Institute of Technology, Boston, Massachusetts, US, 2023

Perfect sampling from spatial mixing

Online Talk, Probability, Information Theory and Computing Workshop, TU Dortmund, Germany, 2023

Towards derandomising Markov chain Monte Carlo

Seminar Talk, University of Science and Technology of China, Hefei, China, 2023

A simple polynomial-time approximation algorithm for the total variation distance between two product distributions

CFCS Youth Forum, Peking University, Beijing, China, 2023

Swiss Winter School on Theoretical Computer Science, Zinal, Switzerland, 2023

A Markov chain approach to the sampling Lovász local lemma.

BARC Talk, University of Copenhagen, Copenhagen, Denmark, 2023

DIMAP Seminar, University of Warwick, Coventry, UK, 2022

Optimal mixing for two-state anti-ferromagnetic spin systems.

FOCS, Denver, Colorado, US, 2022

Field dynamics: a new tool to boost mixing results.

UCSB Summer School, UC Santa Barbara, Santa Barbara, California, US, 2022

Algorithms and Complexity Theory Seminar, University of Oxford, Oxford, UK, 2022

Rapid mixing of Glauber dynamics via spectral independence for all degrees.

HALG, London School of Economics and Political Science, London, UK, 2022

FOCS, online, 2022

IFCS Lab Lunch, University of Edinburgh, Edinburgh, UK, 2022

Chengdu Algorithm & Logic Seminar, University of Electronic Science and Technology of China, Chengdu, China, 2021

Rapid mixing from spectral independence beyond the Boolean domain.

SODA, online, 2021

Distributed Metropolis sampler with optimal parallelism.

SODA, online, 2021

Fast sampling and counting k -SAT solutions in the local lemma regime.

SIGMA Seminar, Institute of Computing Technology, Chinese Academy of Sciences, online, 2020

STOC, online, 2020

IJTCS, online, 2020

Dynamic sampling from graphical models

Nanjing Theory Day, Nanjing University, Nanjing, China, 2019

STOC, Phoenix, Arizona, US, 2019

NCTCS, Lanzhou University, Lanzhou, China, 2019

Microsoft PhD Summit, Microsoft Research Redmond, Redmond, Washington, US, 2019

Local distributed sampling

Microsoft Research Asia, Beijing, China, 2018

Service

Chair of the ACM Hong Kong Chapter (2026–present).

Program Committee for RANDOM 2024.

External reviewer for STOC (2021, 2022, 2023, 2024, 2026), SODA (2022, 2024, 2026), FOCS (2026), RANDOM (2022, 2023, 2025), ISAAC (2023), JACM (2024), SICOMP (2022), IPL (2025).

Organizer (with Xiaoming Sun, Jialin Zhang and Zhijie Zhang) of CCF TCS PhD Forum 2020.